Bayesian Modelling: Gaussian Modelling Lab

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Lab: Gaussian modelling

- Compute the posterior, and the posterior mean, for the midge wing data with unknown σ^2.
- Carry out a sensitivity analysis by varying m_0 and M_0 and comparing results.

Lab: Bayesian linear regression

- Now consider the `MASS::Boston` data set.
- The variables of interest are
  - `medv` (median value of owner-occupied homes in 1000 USD),
  - `age`: the proportion of owner-occupied units built prior to 1940, and
  - `lstat`: lower status frequency in the population (percent).
- Fit a Bayesian posterior distribution to the model

$$\texttt{medv} \sim \texttt{lstat} + \texttt{age}.$$

- Perform a sensitivity analysis on any prior parameters you specified.